



Data Visualization for Research

Presenting Results Effectively





The Whats and Whys of Data Visualization





Introduction

Data Visualization is the **graphical representation of data**. It involves using charts, graphs, maps, and other visual elements to reveal patterns, trends, outliers, and insights more effectively than raw numbers.

There are two different approaches to data visualization:

- **Exploratory:** Your purpose for visualization is the exploration and understanding of trends in your data, either for the purpose of data analysis or the preparation of it.
 - **Explanatory:** Your purpose for visualization is to clearly and effectively communicate something about your data to a wider audience.
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Why Visualize Data?

01 Simplify Complex Data
It turns large and complicated data into visual formats like charts and graphs, making the information easier to understand.

02 Reveals Patterns and Trends
It helps identify trends, relationships, and patterns that are not easily seen in raw data or tables.

03 Improve Communication
It makes it easier to explain data insights to others, especially those who may not be familiar with the technical details.





Limitations

Oversimplification Risk

Important context or nuances may be lost when data is reduced to visual form.

Bias & Misleading Design

Truncated axes, cherry-picked data, or misleading chart types can distort interpretation.

Cognitive Overload

Complex or cluttered visuals can confuse more than clarify.

Data Quality Dependency

Poor or incomplete data leads to misleading visuals, no matter how well designed.



Tools for Visualization

- **Programming:** R (ggplot2, Shiny), Python (Matplotlib, Seaborn, Plotly, Altair)
- **Web based tools:** Javascript libraries like D3.js
- **Third Party Apps:** Microsoft Excel, Tableau, Power BI
- **Temporal, Geospatial, and Exhibition tools:** Omeka, StoryMap, Timeline.js, WorldMap, and Carto





What makes a Good Visualization





Pre-Attentive Attributes

A human can distinguish differences in length, shape, orientation, distances, and color readily without significant processing effort; these are referred to as **pre-attentive attributes**. Compelling graphics take advantage of the attributes.

There are 4 pre-attentive visual properties: **Form**, **Color**, **Spatial Position** and **Movement**. From these four, we are able to derive specific visual features that the visual system can detect instantly.

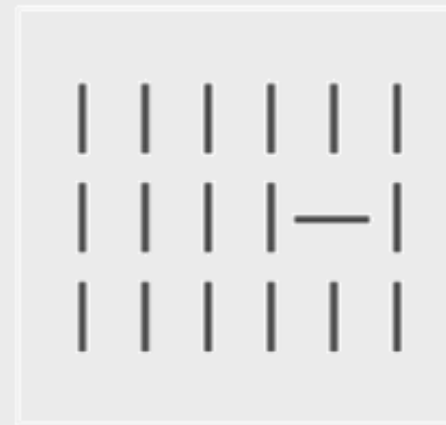




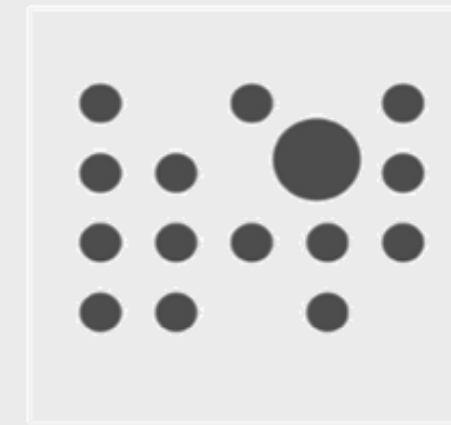
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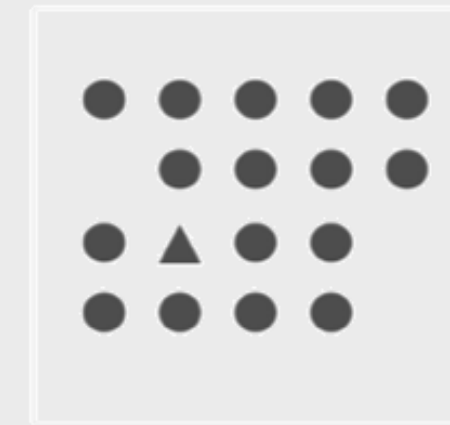
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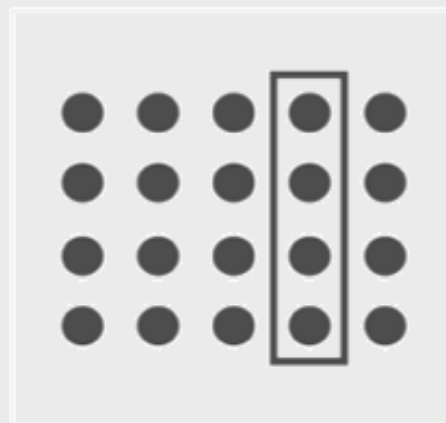
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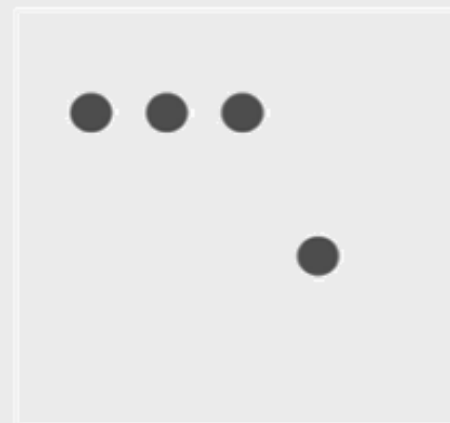
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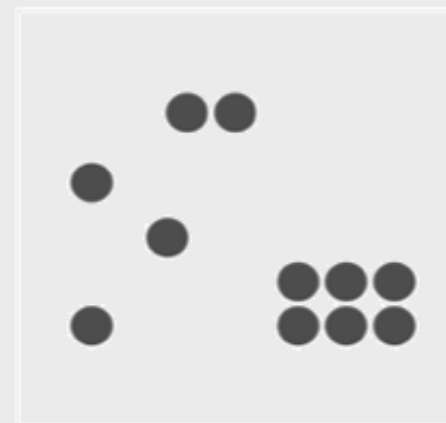
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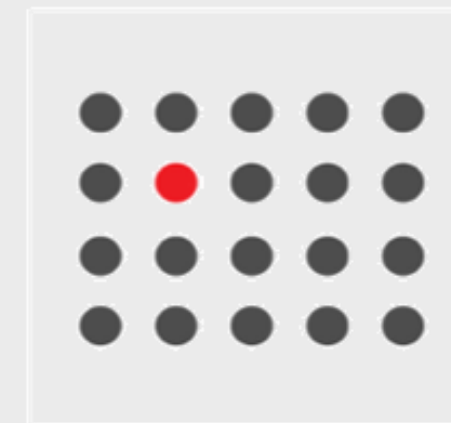
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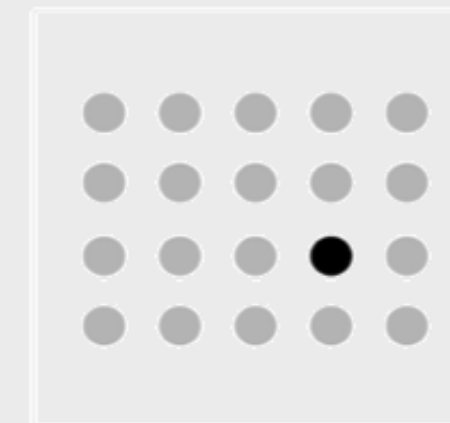
Position



Grouping



Color Hue



Color
Intensity

Specific Pre-attentive Visual Features in Data
Visualization



Quantitative Messages

Quantitative Messages describe the type of insight or comparison a user wants to extract from numerical data, guiding how the data should be visualized.

Each message type is best conveyed through a specific kind of graph or chart, which helps highlight the intended relationship or pattern.

There are **8** types of quantitative messages.





Quantitative Messages

01

Time-Series

Shows how a variable changes over time e.g., temperature over 10 years
→ Line chart

02

Ranking

Orders categories by size (ascending or descending) e.g., top-performing sales reps
→ Bar chart

03

Part-to-Whole

Shows proportions of a whole (percentages) e.g., market share by brand
→ Pie chart, bar chart

04

Deviation

Highlights difference from a reference point e.g., budget vs. actual expenses
→ Bar chart





Quantitative Messages

05

Frequency Distribution

Shows how often values occur within defined intervals e.g., stock returns grouped in ranges
→ Histogram, boxplot

07

Nominal Comparison

Compares values across unordered categories e.g., sales by product code
→ Bar chart

06

Correlation

Shows the relationship between two numerical variables e.g., inflation vs. unemployment
→ Scatter plot

08

Geospatial

Displays data across geographic or spatial layouts e.g., population by state or floor
→ Choropleth map, cartogram





Tufte's Principles

Edward Tufte, a pioneer in data visualization, proposed a set of principles to guide the creation of **clear, honest,** and **efficient** visual representations of data.

He argued that graphics should communicate data directly and efficiently, without unnecessary decoration or distraction. Instead of focusing on design flair, Tufte advocated for visuals that induce the viewer to focus on the substance of the data itself.





Tufte's Principles

Tufte states that graphical displays should:

- show the data
 - induce the viewer to think about the substance rather than about methodology, graphic design, the technology of graphic production, or something else
 - avoid distorting what the data has to say
 - present many numbers in a small space
 - make large data sets coherent
 - encourage the eye to compare different pieces of data
 - reveal the data at several levels of detail, from a broad overview to the fine structure
 - serve a reasonably clear purpose: description, exploration, tabulation, or decoration
 - be closely integrated with the statistical and verbal descriptions of a data set.
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THANK YOU

